LED Challenge 4: Using a Photoresistor.

Purpose: To include a photoresistor that uses light detection to operate an LED.

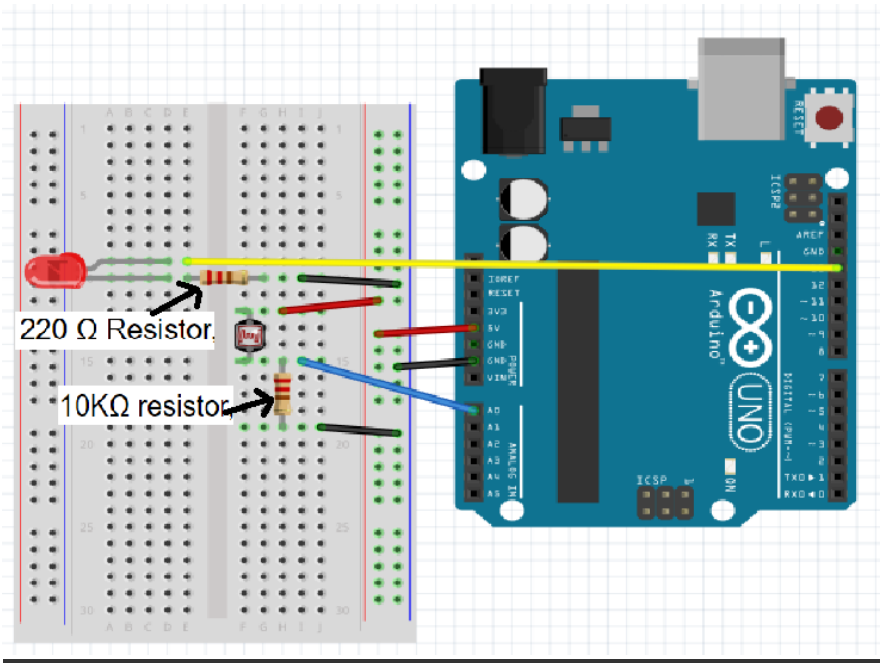
Materials: Computer, Arduino Uno Board, USB cable, Breadboard, 1 LED, 1 - 220 Ω Resistor, 1 - 10KΩ resistor, 7 Jumper Wires, 1 Photoresistor

New Components:
Photoresistor. Photoresistors are light sensitive, variable resistors. As more light shines on the sensor's head, the resistance between its two terminals decreases. Therefore, photoresistors can act as a switch that uses light detection to turn loads (such as light) on and off.



Procedure:

1. Wire the project using the following schematic.



2. You can create your code on the **TinkerCAD website**

3. Below are the blocks that you will need, once complete click on the “Blocks and Text” option to see the text to copy into the **Arduino IDE**.

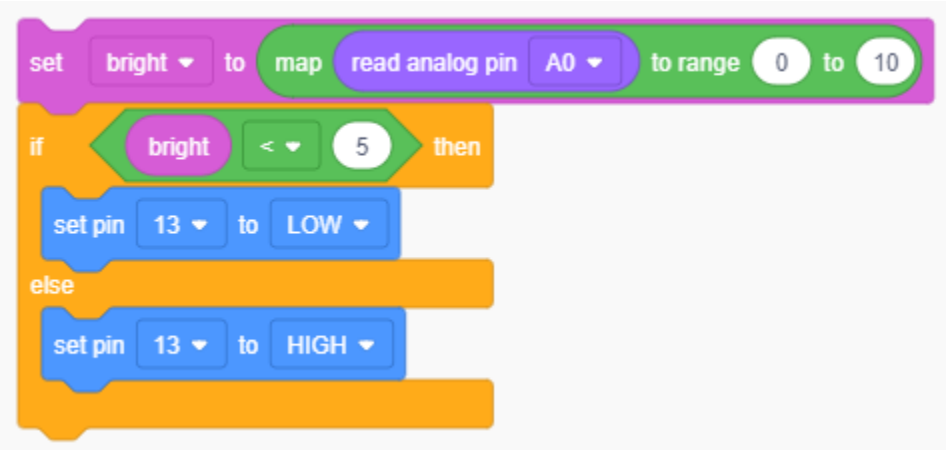
4.

   a. First, we need set up a variable. These are like X and Y you use in math class. **Variables** are used to store information to be referenced and manipulated in a computer program. They also provide a way of labeling data with a descriptive name, so our programs can be understood more clearly by the reader and ourselves. It is helpful to think of variables as containers that hold information. Their sole purpose is to label and store data in memory. This data can then be used throughout your program. We will create the integer variable “bright”.

   b. We are also going to use the “**map**” function (FIND in the MATH blocks). Its re-maps a number from one range to another.

      In this case, since we are using **analogRead** (which produces a result somewhere between 0 and 1023), the value read will be *mapped* to a value between the range given (in this case 0 to 10).

      an **analogRead** of 0 would map to 0
      an **analogRead** of 1023 would map to 10
      and in-between values would map to the proportional position in the new range. (a value of 512 would map to 5)



*****SAVE your sketch to your Robotics/Code Folder that you created*****

Now select the “Arduino” tab you will see the following code.

```
int bright = 0;

void setup()
{
  pinMode(A0, INPUT);
  pinMode(13, OUTPUT);
}

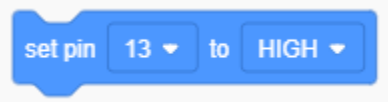
void loop()
{
  bright = map(analogRead(A0), 0, 1023, 0, 10);
  if (bright < 5) {
    digitalWrite(13, LOW);
  } else {
    digitalWrite(13, HIGH);
  }
  delay(10); // Delay a little bit to improve simulation performance
}
```

6. Upload your program to the Arduino. Make sure the Arduino Uno board is hooked up using the supplied USB cord to your computer and click the green "upload" button.

6. Your program is now loaded. Observe your LED and complete your worksheet titled **LED Challenge 4: Using a Photoresistor.**

Troubleshooting: If your circuit is not working!

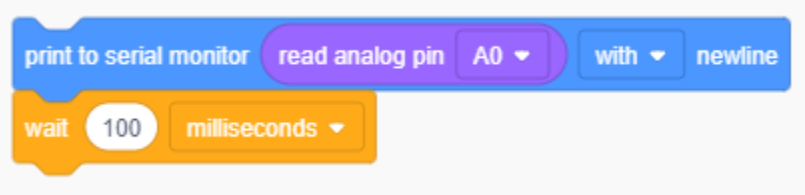
1. Double check your code! Did you put in every block? Are the settings correct?
2. Refresh the website and reload the code.
3. The next step is to check your circuit!
    a. Check your wiring! Is it the same as the circuit diagram? Does the circuit go from the positive numbered pin to the ground?
    b. Next check each component. Check the led, is the long positive arm connected by wires to the pins?
    c. It may be wise to test the led by turning them it on. Open a new blocklyduino window and create this test code. The block should look like this:



    d. LED still not working? Next inserting the LED directly into pin 13 and the adjacent ground. Does it light when you try this test code?

e. Next check the wires in the circuit. This can be done by connecting the LED to pin 13 only then using the wire to ground the short pin. Do not connect the pin to the ground directly. (Shorting the wires) This will damage the UNO.

f. You can also check the photoresistor by checking what signal is being sent to the pin using the serial monitor. With code like this:



Do the values change when you cover the light sensor?

LED Challenge 4: Using a Photoresistor.

1. What does the LED do when you block the light to the photoresistor? Why?

2. What is the purpose a photoresistor? List some real-life applications where a photoresistor is used or an application where one would be useful.

Challenges:

3. Using your coding experience from the previous challenges, make your LED flash at 2 second intervals when it is dark. Test your code by uploading it to your Arduino and observing.

4. Go back to your TinkerCAD and change your code so that your light flashes when there is light and off when there is no light. Test your code by uploading it to your Arduino and observing.

5. What are some real-life applications where photoresistors could be used?

With funding from Canada

<SaskCode>

SIEC Saskatoon Industry Education Council